

PROBLEM STATEMENT & OBJECTIVES

Autonomous Drone Navigation • ROS 2-based GPS-independent MVP

PROJECT PROBLEM

Small autonomous drones commonly depend on GNSS/GPS for global position. In indoor, obstructed or otherwise GNSS-denied environments, that source may be unavailable or unreliable. An IMU provides high-rate motion information, but integrated inertial measurements accumulate position drift. The MVP therefore focuses on onboard sensing, local state estimation, simple goal navigation and obstacle safety in a compact ROS 2 architecture.

MVP QUESTION

How can a small autonomous drone navigate from a defined start position toward a destination using onboard sensor data, without relying on GPS, while responding safely to a simple obstacle?

Project positioning

This is intentionally a ROS course MVP, not a production-ready or research-grade GPS-denied flight system. Full outdoor autonomy, 3-D SLAM and advanced perception are outside the current scope.

1. PROBLEM CONTEXT

Why the MVP problem exists

Manual control: requires continuous pilot attention and does not provide repeatable autonomous trajectories.

GPS dependence: GNSS may be unavailable or unreliable in constrained environments.

IMU drift: integrated motion measurements accumulate error over time.

Obstacle response: localization alone does not guarantee collision avoidance.

Complexity: research systems can require SLAM, LiDAR, UWB, cameras or powerful computing.

MVP response: combine simple onboard sensing, local state estimation, goal navigation, range-based obstacle monitoring and ROS 2 decision logic.

Core engineering problem

Integrate sensing, local state estimation, goal-directed navigation and obstacle safety into a small ROS 2 system that can actually be implemented and demonstrated as an MVP.

2. FORMAL PROBLEM STATEMENT

Source-grounded formulation

Conventional small-drone autonomy often depends on GNSS/GPS for global position. In indoor, urban-obstructed, underground, or otherwise GNSS-denied environments, that position source can be unavailable or unreliable. An IMU provides high-rate motion information but its integrated position estimate accumulates drift. Therefore, a practical autonomous drone needs an onboard state-estimation pipeline that combines IMU data with complementary measurements, together with local obstacle sensing and a navigation/controller stack.

MVP reformulation

For the ROS course, reduce the problem to a controlled local-navigation scenario: demonstrate that a drone can move from START → GOAL using onboard sensing, maintain a usable local state estimate, detect a simple obstacle and respond safely without GPS as the core navigation source.

Engineering gap

The technologies are established. The MVP gap is practical integration: a compact ROS 2 system with clearly separated sensing, estimation, navigation and safety functions, plus an explicit response when sensor confidence or obstacle safety becomes unacceptable.

3. AIM

One clear aim for Project Review-I

To design and develop a ROS 2-based autonomous drone navigation MVP that can travel from a defined start position toward a destination, estimate local motion using onboard sensing without GPS as the navigation source, detect a simple obstacle, and generate safe navigation commands.

MVP success concept

START → LOCAL STATE → GOAL NAVIGATION → OBSTACLE RESPONSE → GOAL

4. SPECIFIC OBJECTIVES

Objectives mapped directly to implementable MVP components

01	ROS 2 integration	Build a modular ROS 2 node architecture connecting sensing, state estimation, navigation, obstacle monitoring and flight commands.
02	Sensor acquisition	Acquire and process IMU data together with a complementary local-motion/range measurement.
03	Local state estimation	Estimate usable local pose/velocity without GPS as the core navigation source.
04	Goal navigation	Accept a destination or waypoint and generate simple movement commands toward it.
05	Obstacle detection	Detect an obstacle inside the local flight corridor using range sensing.
06	Adaptive safety	Change behavior according to sensor confidence and obstacle state: move, slow, stop or safe-hold.
07	PX4 interface	Exchange navigation commands and flight state with a PX4-compatible flight controller where feasible.
08	MVP validation	Demonstrate the complete pipeline through controlled navigation and obstacle-response scenarios.

5. MVP BOUNDARIES & VALIDATION

Keep the project achievable and measurable

IN SCOPE: ROS 2 nodes • IMU + simple complementary sensing • local state estimation • START → GOAL navigation • simple range-based obstacle response • confidence-aware safe states • PX4 interface where feasible.

OUT OF SCOPE: full 3-D SLAM • dynamic obstacle prediction • reinforcement learning • long-range outdoor GPS-denied navigation • research-grade multi-sensor localization • production-ready autonomy.

Validation criteria

- Goal reached without manual takeover in the controlled scenario.
- Obstacle detected inside the configured safety zone.
- System changes behavior appropriately: move / slow / stop / safe-hold.
- ROS 2 topics and interfaces operate correctly.
- Invalid or low-confidence state produces a safe response.

These are validation criteria, not experimental results. No achieved performance should be claimed before testing.

6. FINAL REVIEW-I SUMMARY

The story to present to the faculty

PROBLEM

GPS may be unavailable; IMU-only position drifts; obstacle avoidance requires explicit sensing and decision logic.



MVP

A compact ROS 2 system that estimates local motion, navigates toward a goal, monitors obstacles and changes behavior according to confidence and safety.



OBJECTIVES

Build the ROS 2 pipeline, process sensors, estimate local state, navigate to a goal, detect obstacles, adapt safely and interface with PX4.



DEMONSTRATION

START → navigate → obstacle → respond safely → continue → GOAL.

Recommended one-line project statement

A ROS 2-based MVP demonstrating confidence-aware local autonomous drone navigation without relying on GPS as the core navigation source.

Source basis: DA-1 Complete Project Material, DA-1 Presentation and DA-1 Work Distribution documents supplied for the project.